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NJSD

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JOURNAL INFORMATION

The *Nepal Journal of Statistics and Data Sciences (NJSD)* is a peer-reviewed scholarly journal dedicated to the advancement of statistics, data science, quantitative methods, and their interdisciplinary applications.

The journal welcomes original research, methodological contributions, empirical studies, review articles, and applications of statistical and data-science methods across diverse fields.

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FROM THE EDITOR-IN-CHIEF

It is with great pleasure that we are presenting the inaugural publication of the *Nepal Journal of Statistics and Data Sciences (NJSD)*, the first journal publication of the **Nepal Statistical Society (NSS)**. This issue marks an important milestone not only for the Society but also for the growing community of statisticians, data scientists, researchers, academics, and practitioners in Nepal.

The establishment of the *Nepal Statistical Society* reflects a collective commitment to promoting the study, teaching, application, and advancement of statistics and data science. The publication of NJSD represents a natural extension of this mission: to create a scholarly platform through which researchers can share methodological innovations, empirical evidence, and applications of statistical and data-science techniques to contemporary problems.

This inaugural issue brings together eight original research contributions covering a diverse range of topics, including structural equation modeling, forecasting, artificial intelligence, GIS and IoT, natural calamities, economic modeling, acceptance sampling, and statistical methodology. The diversity of these contributions reflects the increasingly interdisciplinary nature of statistics and data science and demonstrates how quantitative methods can contribute to understanding problems across education, banking, economics, industry, disaster management, and other fields.

Statistics has traditionally served as a foundation for scientific inquiry by providing systematic approaches to collecting, analyzing, interpreting, and communicating evidence. Today, however, the statistical landscape is changing rapidly. The increasing availability of digital data, advances in computing, artificial intelligence, machine learning, geographic information systems, and Internet of Things technologies have created new opportunities for quantitative research. At the same time, these developments have created new responsibilities concerning data quality, methodological rigor, transparency, reproducibility, privacy, and ethical use of data.

In this changing environment, the role of statistics is not diminished; rather, it becomes even more important. Statistical reasoning provides the principles necessary to distinguish meaningful patterns from random variation, association from causation, prediction from explanation, and information from evidence.

The papers included in this issue illustrate this broader role of statistics.

The contributions employing **structural equation modeling** demonstrate the value of statistical techniques for investigating relationships among complex constructs that cannot always be observed directly. Applications in higher education and the banking sector show how quantitative evidence can inform institutional improvement and organizational decision-making.

The study employing **Box–Jenkins ARIMA modeling for Nepal’s nominal GDP** highlights the continuing importance of statistical forecasting for economic analysis. Reliable forecasts can provide useful information for economic planning, policy formulation, resource allocation, and the assessment of future economic conditions.

The application of **artificial intelligence, GIS, and IoT technologies** to industrial productivity demonstrates another emerging direction for quantitative research. The integration of statistical thinking with computational and digital technologies offers considerable potential for developing more responsive and intelligent decision-support systems.

Research examining **deaths associated with natural calamities using Nepal’s census data** illustrates the social value of statistical analysis. Nepal faces substantial exposure to earthquakes, floods, landslides, and other hazards. Systematic analysis of population and demographic data can help identify patterns of vulnerability and contribute to evidence-based disaster preparedness and risk-reduction policies.

Similarly, research concerning **minimum wage determination, economic growth, and inflation** demonstrates how quantitative modeling can contribute to important socioeconomic policy questions. Statistical evidence can help policymakers move beyond assumptions and anecdotal observations toward more systematic assessments of economic relationships.

The work on **empirical Bayesian acceptance sampling** highlights the practical contribution of statistical methodology to quality assurance and decision-making, while the review of the **history,**

methods, and applications of structural equation modeling provides methodological context that can support researchers in appropriately applying advanced statistical techniques.

Although these studies differ substantially in subject matter and methodology, they share a common foundation: the use of quantitative evidence to understand real-world phenomena and support better decisions.

This is precisely the kind of research culture that the *Nepal Statistical Society* seeks to foster.

The launch of NJSD is therefore more than the publication of another academic journal. It represents the beginning of an institutional platform for bringing together researchers and practitioners who share an interest in statistics and data science. We hope the journal will encourage researchers from Nepal and beyond to contribute rigorous theoretical, methodological, empirical, and interdisciplinary research.

Nepal is experiencing rapid growth in the generation and use of data across government, business, education, development, health, finance, environment, and technology. However, the availability of data alone does not guarantee better decisions. The transformation of data into reliable knowledge requires appropriate research design, sound statistical methods, careful interpretation, and effective communication of evidence.

The Society therefore has an important role to play in strengthening statistical capacity and promoting responsible data practices. NJSD will contribute to this broader objective by providing a forum for scholarly communication and by encouraging research that addresses both national and international challenges.

We extend our sincere appreciation to all authors who have contributed their research to this inaugural issue. We are equally grateful to the reviewers for their time, expertise, and constructive assessments, and to the editorial team and members of the Nepal Statistical Society whose efforts have made this first publication possible.

The publication of this issue is a beginning. Much remains to be done to strengthen statistical research, education, professional practice, and data science in Nepal. We invite researchers, academics, practitioners, and students to participate in this journey by sharing their knowledge, questioning established approaches, developing new methods, and applying quantitative evidence to the challenges facing our society.

As the *Nepal Statistical Society* takes this first step in scholarly publishing through the *Nepal Journal of Statistics and Data Sciences*, we look forward to building a vibrant and inclusive community dedicated to the advancement of statistics and data science.

Bijay Lal Pradhan, PhD

Editor-in-Chief

Nepal Journal of Statistics and Data Sciences

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CONTENTS

Research Articles

Perceived Service Quality and Students' Satisfaction in a Higher Education Institution: A Structural Equation Modeling Approach

Hari Prasad Upadhyay, Bijay Lal Pradhan, Ram Prasad Khatiwada 1–9

Modeling and Forecasting Nepal's Nominal GDP with an Updated Box Jenkins ARIMA Analysis

Udaya Bahadur Bohara, Bijay Lal Pradhan 10–18

AI-Driven Productivity Impact Analysis Using GIS and IoT Data in Sugar Industries: A Pilot Study

Amol Chavan, Santosh Parakh 19–24

Natural Calamity Deaths in Nepal: Analysis of the 2021 National Population and Housing Census

Jibesh Acharya, Srisa Rijal, Divya Shakya 25–32

Modeling Minimum Wage Determination in Nepal: The Role of Economic Growth and Inflation

Ishwar Kumar Shrestha, Nabaraj Poudyal, Eak Raj Kadariya 33–38

Development of an Empirical Bayesian Single-Attribute Acceptance Sampling Plan for Food Quality using a Beta–Binomial Model

Ram Prasad Khatiwada 39–49

Structural Equation Modeling: A Review of Its History, Methods, and Applications

Bijay Lal Pradhan 50–58

Structural Modeling of Employee Performance: The Roles of Capacity Building and Supervisor Support in Nepalese Commercial Banks

Asfaq Ahmad, Govind Jnawali 59–66

Volume 1, Issue 1, 2026

Nepal Journal of Statistics and Data Sciences
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